

DAYANANDA SAGAR COLLEGE OF ENGINEERING

**Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078
Department of Artificial Intelligence and Machine Learning**



INTERNSHIP REPORT

ON

“PERFORMANCE PREDICTION OF ANTENNA CONTROL SERVO SYSTEM (ACSS) BASED ON LSTM NETWORK”

Submitted in partial fulfillment for the award of degree(20AI8ICINT)

BACHELOR OF ENGINEERING IN

Department of Artificial Intelligence and Machine Learning

Submitted by:

Nishank K S - 1DS20AI036

Conducted at



**Indian Space Research Organization
Telemetry, Tracking and Command Network**

**Department of Artificial Intelligence and Machine Learning
Dayananda Sagar College of Engineering Bangalore-78**

DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078
Department of Artificial Intelligence and Machine Learning



CERTIFICATE

This is to certify that the Internship titled “**Performance Prediction of Antenna Control Servo System (ACSS) based on LSTM Network**” carried out by **Mr NISHANK K S [1DS20AI036]**, a bonafide student of Dayananda Sagar College of Engineering, in partial fulfillment for the award of **Bachelor of Engineering, in Artificial Intelligence and Machine Learning** during the year 2023-2024. It is certified that all corrections/suggestions indicated have been incorporated in the report.

The internship report has been approved as it satisfies the academic requirements in respect course Internship (20AI8ICINT)

Signature of Reporting Head
Parvathi Devi M
Manager
ISTRAC, ISRO

Signature of Internship Coordinator
Prof. Kavya D N
Dept. of AI & ML
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Signature of HoD
Dr. Vindhya P Malagi
Dept. of AI & ML
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External Viva:

Name of the Examiner

Signature with Date

1) _____

2) _____

DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078

Department of Artificial Intelligence and Machine Learning



DECLARATION

I, **Nishank K S**, final year student of Artificial Intelligence and Machine Learning, Dayananda Sagar College of Engineering, Bangalore - 560 078, declare that the Internship has been successfully completed, in **Indian Space Research Organization**. This report is submitted in partial fulfillment of the requirements for the award of Bachelor's degree in Artificial Intelligence and Machine Learning, during the academic year 2023- 2024.

Date:

USN: 1DS20AI036

Place: Bangalore

NAME: Nishank K S

OFFER LETTER

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NO.ISTRAC/HRD/P2/2024 (03)

24 January, 2024

The HOD,
Dept. of Artificial Intelligence and Machine Learning,
Dayananda Sagar College of Engineering,
Bangalore.

Name of the Student:
Mr. Nishank K S

Sir/Madam,

Sub: Allotment of Project Work for your students at ISTRAC/ISRO -Reg.
Ref: Letter(DSCE/AI/ML/2023-24/5) Dated: 17.01.2024

In connection with the above subject and reference, your request for allotment of Project Work at ISTRAC for your students to be carried out during as partial fulfilment for completion of B.E in Artificial Intelligence & Machine Learning is considered favourably. The Project Work will be carried out from 01/02/2024 to 01/05/2024. The students shall possess an aggregate of Min.60% or CGPA 6.32 on a scale of 10.

Enclosed please find a set of forms to be duly filled up by the student and attested/approved by the concerned HOD/Principal. You are requested to advise the student to send the undersigned with all the forms completed in all respects as per the project schedule. You are also requested to advise the student to send, **Bonafide certificate issued by the Principal in original** addressed to **Manager, HRD, ISTRAC**, for the verification of the credentials of the student and a set of the attested copies of all certificates & marks cards with respect to the educational qualifications mentioned in the forms for further processing at ISTRAC, and to produce the copy of Aadhar card at the time of visit.

It may please be noted that the working model in case of hardware project or the source code in case of the software project will not be provided to the student to be taken out of ISTRAC. Mobile phones and any other kind of electronic items are not allowed inside the campus. The project trainees/In-plant trainees/Internship shall not be eligible for any stipend / remuneration / financial assistance.

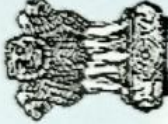
Note: Foreign nationals and NRI students are not allowed inside the campus.

Regards,


(B SANKAR MADASWAMY)

भारतीय अन्तरिक्ष अनुसंधान संगठन (इसरो) INDIAN SPACE RESEARCH ORGANISATION (ISRO)

COMPLETION CERTIFICATE

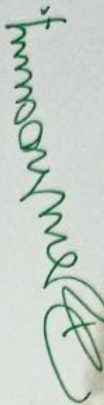


GOVERNMENT OF INDIA
DEPARTMENT OF SPACE
INDIAN SPACE RESEARCH ORGANISATION
ISRO TELEMETRY TRACKING AND COMMAND NETWORK
BANGALORE - 560 058

CERTIFICATE

This is to certify that Sri/Smt/Kum **Nishank Satish**, a student of **Dayananda Sagar College of Engineering, Bangalore** has successfully completed the Project Work towards the partial fulfilment for **B.E (Artificial Intelligence & Machine Learning)** degree from 01/02/2024 to 01/05/2024. He has worked on "PERFORMANCE PREDICTION OF ANTENNA CONTROL SERVO SYSTEM (ACSS) BASED ON LSTM NETWORK" during this period.

M. Parvathi / 01.05.2024
Parvathi Devi M
Scientist/Engineer SF
(Guide)
SDD
ISTRAC/ISRO


B. SANKAR MADASWAMY
MANAGER, HRDD
ISTRAC/ISRO


DATE: 01.05.2024 **HRD NO. 00103/2024(28)**

ACKNOWLEDGEMENT

This Internship is a result of accumulated guidance, direction and support of several important persons. We take this opportunity to express our gratitude to all who have helped us to complete the Internship.

We express our sincere thanks to our Principal **Dr. B G Prasad**, for providing us adequate facilities to undertake this Internship.

We would like to thank **Dr. Vindhya P Malagi**, Head of Department – AI & ML, for providing us an opportunity to carry out Internship and for her valuable guidance and support.

We would like to thank **Mrs. Parvathi Devi M**, ISTRAC, ISRO, for guiding us during the period of internship.

We express our deep and profound gratitude to our Internship Co-ordinator, **Prof. Kavya D N**, Assistant Professor – AI & ML, for her keen interest and encouragement at every step in completing the Internship.

We would like to thank all the faculty members of AI & ML department for the support extended during the course of Internship.

We would like to thank the non-teaching members of the Department of AI & ML, for helping us during the Internship.

Last but not the least, we would like to thank our parents and friends without whose constant help, the completion of Internship would have not been possible.

NISHANK K S [1DS20AI036]

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CHAPTER 1

INTRODUCTION

INTRODUCTION TO EQUIPMENT PERFORMANCE PREDICTION

Equipment performance prediction is a critical aspect of modern asset management strategies, aiming to anticipate and mitigate potential issues with essential machinery and equipment. It involves utilizing various data-driven techniques to forecast the future behavior of critical equipment within an operational environment. This proactive approach enables organizations to anticipate potential failures, optimize maintenance schedules, and minimize downtime, thereby enhancing operational efficiency and reducing costs. Some key advantages of equipment performance prediction are:

1. **Early Detection:** By analyzing historical data and identifying patterns, equipment performance prediction can detect potential failures before they occur, allowing for proactive maintenance interventions.
2. **Cost Savings:** Predictive maintenance helps organizations avoid costly unplanned downtime by addressing issues before they escalate into major problems, reducing repair expenses and operational losses.
3. **Improved Reliability:** By ensuring that equipment is properly maintained and operating optimally, performance prediction enhances overall reliability, leading to increased uptime and productivity.
4. **Enhanced Safety:** Predictive maintenance minimizes the risk of equipment failures that could compromise safety in the workplace, protecting both employees and assets.
5. **Data-Driven Decision Making:** Equipment performance prediction leverages data analytics to provide actionable insights, empowering organizations to make informed decisions about maintenance strategies and resource allocation.

In conclusion, equipment performance prediction offers significant benefits to organizations, including early detection of failures, cost savings, improved reliability, enhanced safety, and data-driven decision-making. By implementing proactive maintenance strategies and leveraging predictive analytics, businesses can optimize their operations, minimize risks, and maintain a competitive edge in today's dynamic environment.

CHAPTER 2

DESCRIPTION OF ORGANIZATION

INDIAN SPACE RESEARCH ORGANIZATION



Introduction

The Indian Space Research Organisation (ISRO) was founded in 1969 with the goal of developing India's space capabilities and using space technology for the country's socio-economic development. India's first satellite, Aryabhata, was launched in 1975 with the help of the Soviet Union. The launch of Aryabhata marked India's entry into the world of space technology.

Since then, ISRO has achieved several milestones in the field of space technology. The organization has launched several satellites for various applications such as communication, remote sensing, navigation, and scientific research. ISRO's first communication satellite, INSAT-1B, was launched in 1983, and its first remote sensing satellite, IRS-1C, was launched in 1995. ISRO's first lunar mission, Chandrayaan-1, was launched in 2008, and its first interplanetary mission, Mars Orbiter Mission, was launched in 2013.

ISRO has also developed several launch vehicles to launch its satellites into space. Its first launch vehicle, SLV, was used to launch Aryabhata in 1975. PSLV, or Polar Satellite Launch Vehicle, was developed in the 1990s and is capable of launching satellites into polar orbits. GSLV, or Geosynchronous Satellite Launch Vehicle, is capable of launching heavier satellites into geostationary orbits. LVM III, or Launch Vehicle Mark III, is India's most powerful launch vehicle and has been used for the Chandrayaan-2 and Chandrayaan-3 missions and the upcoming Gaganyaan mission.

ISRO has achieved several major recent achievements. In 2017, it launched a record 104 satellites into space in a single mission. In 2019, it successfully launched the Chandrayaan-2 mission, India's second lunar mission, which aimed to explore the south pole of the moon. Though the mission encountered some technical difficulties, it was still considered a significant

achievement for India's space program. In 2023, ISRO made a victorious comeback by successfully performing soft landing on the south pole of the moon in the Chandrayaan 3 mission. ISRO also made headlines when it launched Aditya-L1, India's first solar mission successfully in 2023.

ISRO has several upcoming missions in its pipeline, including Gaganyaan, India's first manned mission to space, Shukrayaan-1, India's first mission to Venus, and Lunar Polar Exploration Mission, ISRO's fourth lunar mission. These missions will further enhance ISRO's capabilities and pave the way for more advanced space exploration.

ISRO has also been collaborating with other space agencies, such as NASA and ESA, on various space missions. In 2020, ISRO and NASA signed an agreement to collaborate on the NASA-ISRO Synthetic Aperture Radar (NISAR) mission, which aims to study the earth's surface and its changes over time using radar imaging.

In conclusion, ISRO has come a long way since its establishment in 1969 and has made significant contributions to India's space capabilities. With its achievements in space and technology and several upcoming missions, ISRO is poised to continue its growth and achieve new milestones in the field of space technology. The organization's history and recent achievements demonstrate its dedication to developing India's space capabilities and using space technology for the country's socio-economic development.

Founder



Vikram Sarabhai, often hailed as the father of the Indian space program, was a visionary scientist, innovator, and institution-builder. Born on August 12, 1919, in Ahmedabad, India,

Sarabhai pursued his education in the field of physics, earning a doctorate from the University of Cambridge. With a keen interest in harnessing space technology for national development, Sarabhai played a pivotal role in shaping India's space program and laying the foundation for the Indian Space Research Organisation (ISRO).

In 1962, Vikram Sarabhai founded the Indian National Committee for Space Research (INCOSPAR), which later evolved into ISRO. His vision was to utilize space technology for societal benefits, including telecommunications, weather forecasting, resource management, and national security. Under his leadership, ISRO launched its first satellite, Aryabhata, in 1975, marking India's entry into space exploration. Sarabhai's emphasis on indigenous development and collaboration with international partners paved the way for India's successful space missions and technological advancements.

During his tenure as chairman of ISRO from 1963 until his untimely death in 1971, Vikram Sarabhai spearheaded several groundbreaking initiatives that propelled India into the realm of space exploration. Under his guidance, ISRO developed a robust infrastructure, including the Thumba Equatorial Rocket Launching Station and the Indian Space Research Organisation Satellite Centre, to support research and development activities. Sarabhai's commitment to innovation and excellence fostered a culture of scientific inquiry and technological innovation within ISRO, leading to significant achievements such as the launch of the Rohini series of satellites and the successful development of the Satellite Instructional Television Experiment (SITE) program. His legacy continues to inspire generations of scientists and engineers to push the boundaries of space exploration and contribute to India's advancement on the global stage.

Chairman



S. Somanath, the current chairman of the Indian Space Research Organisation (ISRO), is a distinguished aerospace engineer and visionary leader in the field of space exploration. With

a career spanning over three decades in ISRO, Somanath has made significant contributions to India's space program and technological advancements. Born and raised in Kerala, India, Somanath pursued his education in aerospace engineering, earning a Bachelor's degree from the Indian Institute of Technology (IIT), Madras, and a Master's degree from the Indian Institute of Science (IISc), Bangalore.

Somanath's journey with ISRO began in 1985 when he joined the Vikram Sarabhai Space Centre (VSSC) as an engineer. Over the years, he held various positions of increasing responsibility within ISRO, contributing to the design, development, and launch of numerous satellites and launch vehicles. Somanath played a crucial role in the successful realization of key projects such as the Polar Satellite Launch Vehicle (PSLV) and the Geosynchronous Satellite Launch Vehicle (GSLV), which have bolstered India's space capabilities and garnered international recognition.

In January 2020, S. Somanath was appointed as the chairman of ISRO, succeeding K. Sivan. As chairman, he has been instrumental in steering ISRO's ambitious initiatives, including the Chandrayaan-2 mission to explore the lunar surface and the ongoing development of the Gaganyaan mission to send Indian astronauts to space. Somanath's leadership and strategic vision have further strengthened ISRO's position as a global leader in space exploration and technology innovation. His dedication to advancing India's space program and fostering collaboration with international partners continue to inspire the next generation of scientists and engineers to pursue excellence in space science and technology.

ABOUT THE COMPANY

About ISRO Telemetry, Tracking and Command Network

ISRO Telemetry, Tracking and Command Network (ISTRAC), Bengaluru is entrusted with the major responsibility to provide tracking support for all the satellite and launch vehicle missions of ISRO. The major objectives of the centre are: carrying out mission operations of all operational remote sensing and scientific satellites, providing Telemetry, Tracking and Command (TTC) services from launch vehicle lift-off till injection of satellite into orbit and to estimate its preliminary orbit in space and hardware and software developmental activities that enhance the capabilities of ISTRAC for providing flawless TTC and Mission Operations services. Towards, these objectives, ISTRAC has established a network of ground stations at Bengaluru, Lucknow, Mauritius, Sriharikota, Port Blair, Thiruvananthapuram, Brunei, Biak (Indonesia) and the Deep Space Network Stations.

In keeping with its long-established TTC support responsibility, ISTRAC has also been mandated to provide space operations support for Deep Space Missions of ISRO, undertake development of radar systems for launch vehicle tracking and meteorological applications, establish and operationalise the ground segment for Indian Regional Navigational Satellite System, provide Search & Rescue and Disaster Management Services and support space based services like telemedicine, Village Resource Centre (VRC) and tele-education.

VISION STATEMENT

To be a world-class center for telemetry, tracking, and command operations, supporting India's space exploration goals by providing state-of-the-art ground support services for satellite and launch vehicle missions.

MISSION STATEMENT

To provide reliable and efficient telemetry, tracking, and command support for satellite and launch vehicle missions, ensuring successful communication and data transfer between ground stations and space missions, while developing and maintaining ground infrastructure and facilities necessary for space mission operations, and collaborating with national and international space agencies to enhance space exploration capabilities.

CHAPTER 3

SCOPE OF WORK

During the internship tenure, I was involved in diverse projects encompassing equipment performance prediction of Antenna Control Servo System (ACSS), visibility file prediction of X1 satellite (name changed), and trend analysis of Antenna Control Unit (ACU) parameters. These projects offered valuable opportunities to delve into various aspects of telemetry data analysis and predictive modeling, contributing to a comprehensive understanding of satellite operations and performance optimization.

1. Equipment Performance Monitoring of ACSS:

Objective: The objective of this project was to develop a predictive maintenance solution using LSTM networks for equipment performance monitoring, aiming to enhance reliability and reduce downtime.

Scope: The scope involved analyzing telemetry data collected from ACSS, preprocessing it to remove noise and anomalies, training LSTM models to predict equipment performance, and evaluating their performance metrics.

Methodology: Utilizing historical telemetry data, LSTM models were trained to recognize patterns indicative of equipment degradation or failure, enabling proactive maintenance interventions.

Results: The project achieved accurate predictions of equipment performance, enabling preemptive maintenance actions and contributing to improved operational efficiency.

2. Visibility File Prediction of Satellite:

Objective: This project aimed to develop a satellite visibility prediction system using LSTM/RNN networks to optimize ground station operations and enhance communication planning.

Scope: Preprocessing satellite visibility data, training LSTM/RNN models, and evaluating their effectiveness in forecasting future visibility parameters were within the project's scope.

Methodology: Historical satellite visibility data was utilized to train LSTM and RNN models capable of predicting satellite visibility patterns, thereby facilitating informed decision-making for ground station scheduling.

Results: The project delivered accurate predictions of satellite visibility, empowering ground station operators with timely information for efficient communication planning and resource allocation.

3. Trend Analysis of ACU Parameters:

Objective: This project focused on conducting trend analysis of crucial parameters in satellite ground stations to monitor performance and optimize operations.

Scope: Analyzing historical data, identifying trends in ACU parameters, and deriving actionable insights for improving station efficiency constituted the project's scope.

Methodology: Statistical techniques and data visualization tools were employed to analyze trends in various ACU parameters over time, enabling the identification of performance patterns and anomalies.

Results: Actionable recommendations were provided for optimizing ground station operations, enhancing performance, and ensuring minimal downtime, based on the insights derived from trend analysis.

In conclusion, the internship projects provided a holistic understanding of telemetry data analysis, predictive modeling, and operational optimization within satellite ground stations. The outcomes of these projects not only contributed to enhancing operational efficiency but also underscored the importance of proactive maintenance and informed decision-making in satellite operations.

CHAPTER 4

LEARNING OBJECTIVES

1. Mastery of Machine Learning and Deep Learning Techniques:

Develop mastery in machine learning and deep learning methodologies among interns, highlighting their critical role in addressing complex challenges within satellite technology. Interns are expected to independently acquire proficiency in a diverse range of machine learning algorithms, including supervised and unsupervised methods, as well as advanced deep learning architectures such as convolutional neural networks (CNNs) and recurrent neural networks (RNNs). Through self-directed study and practical application, interns are expected to demonstrate competency in model development, evaluation, and deployment, enabling them to build predictive models for analyzing satellite telemetry data, forecasting equipment performance, and predicting satellite visibility effectively.

2. Advanced Data Management Skills:

The company outlines the objective for interns to develop advanced data handling and visualization skills necessary for effectively managing and visualizing large-scale satellite telemetry datasets. Interns are tasked with independently acquiring expertise in advanced data preprocessing techniques, including standardization, normalization, and feature engineering, to ensure data quality and suitability for modeling tasks. Similarly, interns are expected to explore various data visualization tools and libraries, such as Matplotlib, Seaborn, and Plotly, to create insightful visual representations of complex data. Through self-directed learning and hands-on practice, interns are expected to uncover meaningful insights, detect patterns, and communicate findings effectively through impactful visualizations, thereby enhancing their contribution to data-driven decision-making processes within the organization.

3. Understanding of Satellite and Launch Vehicle Operations:

The company sets the objective for interns to gain a comprehensive understanding of satellite and launch vehicle operations, essential for navigating the complexities of space missions. Interns are expected to independently explore satellite systems, launch vehicle configurations, and mission planning processes through self-study and research. By immersing themselves in these topics, interns will develop insights into satellite orbits, launch procedures, and mission lifecycles, equipping them with valuable knowledge of the operational aspects of space technology. Through self-directed learning and independent exploration, interns are expected to

demonstrate understanding and proficiency in satellite and launch vehicle operations, contributing to their overall competency in the field of space technology.

4. Proficiency in End-to-End System Integration and Deployment:

The company sets the objective for interns to achieve proficiency in end-to-end system integration and deployment, essential for orchestrating comprehensive solutions for complex technological systems. Interns are tasked with independently developing skills in system architecture design, testing, and deployment through self-directed learning and practical experience. By engaging in hands-on projects and collaborative tasks, interns will acquire proficiency in integrating disparate components of satellite systems, including hardware, software, and communication interfaces, to ensure seamless functionality and interoperability. Through self-directed learning and independent exploration, interns are expected to demonstrate competency in system integration and deployment, contributing to the development and implementation of satellite systems and ground station infrastructure.

5. Application of Artificial Intelligence in Space Technology Research:

The company emphasizes the importance of interns possessing knowledge and skills in applying artificial intelligence (AI) to space technology research. Interns are tasked with independently gaining insights into AI-driven techniques for satellite data analysis, mission planning, and autonomous spacecraft operations through self-study and exploration. By immersing themselves in AI research methodologies and their applications in advancing space technology, interns will develop a nuanced understanding of the role of AI in the field of space exploration. Through self-directed learning and research endeavors, interns are expected to demonstrate proficiency in applying AI techniques to solve complex space-related challenges, positioning themselves as innovators in the space industry.

CHAPTER 5

CHALLENGES AND SOLUTION

1. Equipment Performance Monitoring of ACSS:

Challenges:

- **Data Variability:** The telemetry data collected from Antenna Control Servo Systems (ACSS) may exhibit significant variability due to factors such as environmental conditions, satellite movement, and operational dynamics, posing challenges in identifying consistent patterns and trends.
- **Feature Selection:** Selecting relevant features from the telemetry data for equipment performance prediction requires careful consideration of factors influencing ACSS behavior. Choosing the right set of features that adequately represent the system's performance while minimizing noise and redundancy is crucial but challenging.
- **Data Preprocessing:** Preprocessing the telemetry data involves tasks such as cleaning, normalization, and handling missing values. Dealing with outliers, incomplete records, and data inconsistencies can be challenging and may require sophisticated techniques to ensure data quality and integrity.
- **Model Complexity:** Developing accurate prediction models for ACSS performance often involves dealing with complex relationships and nonlinear dynamics. Choosing appropriate modeling techniques and architectures that can capture these complexities effectively while avoiding overfitting is essential but challenging.
- **Performance Evaluation:** Evaluating the performance of prediction models for ACSS requires robust metrics and validation procedures. Ensuring the reliability and generalization capability of the models across different scenarios and operational conditions presents challenges in model validation and assessment.

Solution:

When dealing with machines requiring periodic maintenance, there are typically three outcomes: over-maintenance, under-maintenance, or timely maintenance. Equipment performance prediction becomes invaluable in forecasting current profiles for fixed demands and signalling when maintenance is due. Achieving this entails constructing an equipment performance prediction model, learning from historical data across various demands to forecast current profiles for future satellite passes and detecting performance degradation for timely repairs. Monitoring current provides critical insights into motor condition, including load monitoring and efficiency assessment. High currents may indicate overloading and deviations from expected current values signal inefficiencies, highlighting areas for energy efficiency improvements. To

address these challenges, an equipment performance prediction model emerges as a promising solution for forecasting and detecting performance degradation. A model was developed to predict ground station antenna current profiles for fixed demands, trained on historical data using the servo log of the Bangalore based ground station antenna. Employing techniques like the sliding window method and feature extraction, a deep learning algorithm namely LSTM which is suited for sequence modelling, is modelled. The application of equipment performance prediction enables proactive detection and resolution of potential equipment degradations.

2. Visibility File Prediction of Satellite:

Challenges:

- **Data Sparsity:** Satellite visibility data may be sparse, leading to gaps in the dataset, which can pose challenges for model training and prediction accuracy.
- **Seasonal Variations:** Variations in satellite visibility patterns due to seasonal changes in atmospheric conditions and orbital dynamics can make it difficult for models to generalize effectively.
- **Noise and Anomalies:** Presence of noise, outliers, and anomalies in the visibility data can adversely affect model performance and lead to inaccurate predictions.
- **Dynamic Environmental Factors:** Dynamic environmental factors such as weather conditions, atmospheric interference, and orbital debris can impact satellite visibility and introduce additional complexity to the prediction task.
- **Complex Temporal Dependencies:** Satellite visibility parameters exhibit complex temporal dependencies, making it challenging for models to capture and extrapolate future trends accurately.

Solution:

Satellite visibility prediction plays a pivotal role in the seamless operation of satellite tracking and communication systems. It involves anticipating the periods during which a satellite, such as the X1 satellite in this project, remains visible and trackable by a designated ground station, in this case, Bangalore Station. In our endeavour to address this challenge, the primary objective was to develop a robust model capable of accurately forecasting the satellite's visibility (varies for different satellite) parameters. Traditionally, various methods such as Two-Line Element (TLE) propagation models and empirical algorithms have been employed for satellite visibility prediction. However, this implementation diverges from conventional approaches by focusing on the utilization of machine learning (ML) and deep learning (DL) techniques to tackle the prediction task.

The project's methodology revolved around creating a continuous dataset from the available sporadic data of satellite passes. ML and DL models were then applied to this continuous dataset to learn the intricate temporal patterns and dependencies inherent in satellite visibility data. By leveraging the power DL architectures such as recurrent neural networks (RNNs) and long short-term memory (LSTM) networks, the goal was to develop a predictive model capable of extrapolating future visibility parameters based on historical data.

3. Trend Analysis of ACU Parameters:

Challenges:

- **Data Volume and Variety:** Managing and analyzing the large volume and diverse nature of ACU parameter data collected from multiple satellites presents challenges in data storage, retrieval, and processing.
- **Noise and Anomalies:** Identifying and filtering out noise, outliers, and anomalies in ACU parameter datasets to ensure the accuracy and reliability of trend analysis results.
- **Complexity of Parameters:** Understanding the complex interrelationships between various ACU parameters and their impact on satellite performance requires comprehensive domain knowledge and analytical expertise.
- **Temporal and Spatial Variability:** Accounting for temporal and spatial variability in ACU parameters across different satellite passes and orbits to capture meaningful trends and patterns.
- **Visualization and Interpretation:** Effectively visualizing and interpreting trend analysis results to derive actionable insights and inform decision-making processes within the organization.

Solution:

Trend analysis in satellite ground stations is crucial for continuous monitoring of various parameters, fault detection and optimization. By examining historical data trends over time, operators can monitor key performance metrics such as current drawn, torque demand and various other parameters to identify deviations, detect equipment malfunctions, and optimize operations. This analysis enables proactive management of station performance, facilitating timely interventions, resource allocation, and process improvements to maintain high communication quality and reliability. The project aims to conduct a comprehensive trend analysis leveraging Antenna Control Unit (ACU) servo log data procured from the Bangalore based ground station. This dataset serves as a repository of invaluable information, encapsulating the operational dynamics of the servo tracking numerous satellites. Each satellite contributes its distinctive set of parameters, ranging from 'iElM1', 'AZ_Ang', 'EL_Ang' to 'AzVel', 'ElVel', 'iAzM1', and 'iAzM2', among others. These parameters serve as vital indicators,

offering insights into the intricate workings of satellite communication systems. The objective of this study is the evaluation of parameter performance across diverse satellite passes. This entails examination of crucial metrics such as torque, current draw, and behavioural patterns under varying operational conditions. Of particular interest is the analysis of torque demand and current draw of the servo of the ACU. By scrutinizing these metrics over time, the project aims to discern trends, anomalies, and potential correlations, thereby facilitating informed decision. Moreover, the project places significant emphasis on understanding parameter behaviour under specific conditions, notably ranges of maximum elevation angles and antenna failure events. The variations of parameters during different ranges of elevation angles offers valuable insights into power consumption patterns and operational efficiency. Simultaneously, the identification and analysis of various parameters during antenna failure events, signalled by deviations in parameters such as iElM1 and iElM2, provide critical inputs for fault detection and mitigation strategies.

HARDWARE AND SOFTWARE REQUIREMENTS

Software requirements:

- Operating system: Windows 10 64 bit
- Python: 3.12.0
- Pandas: 2.2.0
- Scikit-Learn: 1.4.2
- Tensorflow: 2.16.1
- Keras: 3.2.1
- Anaconda Distribution: 2023.09
- PyCharm Community Edition: 2024.1

Hardware Requirements:

- Processor: Intel i7
- RAM: 8 GB minimum
- Display: 23 LED Display
- Input Devices: Mouse, Keyboard
- LAN Connection Port: Required
- Internet Fiber Optic Cable: Required

CHAPTER 6

ACHEIVEMENTS AND CONTRIBUTION

1. Equipment Performance Monitoring of ACSS:

Data Pre-processing and Analysis:

The data under analysis comprises the servo log of the Bangalore based ground station Antenna Control Unit (ACU), encompassing 46 distinct sensor features. These features include time-related parameters such as Hours, Minutes, Seconds, alongside Elevation Velocity, Azimuth Velocity, Torque Elevation, Torque Azimuth, Digital to Analog Converter values, among others. Collected data spans numerous passes of 41 different satellites from July 2022 to April 2024, with continuous logging occurring exclusively during satellite tracking sessions, archived in text format across multiple files. The features included Hours (HH), Minutes (MM), Seconds (SS), Milliseconds (MS), Elevation Velocity (ElVel), Azimuth Velocity (AzVel), Elevation Motor 1 Current (iElM1), Elevation Motor 2 Current (iElM2), Azimuth Motor 1 Current (iAzM1), Azimuth Motor 2 Current (iAzM2) etc. Upon compilation and processing of the servo log files, pertinent information was extracted, including satellite pass date, satellite name, and pass duration. Subsequently, calculations were conducted to derive the Day of Year (DOY) and total seconds for each record in the dataset. Figure 1 represents the correlation matrix of the important features in the dataset.

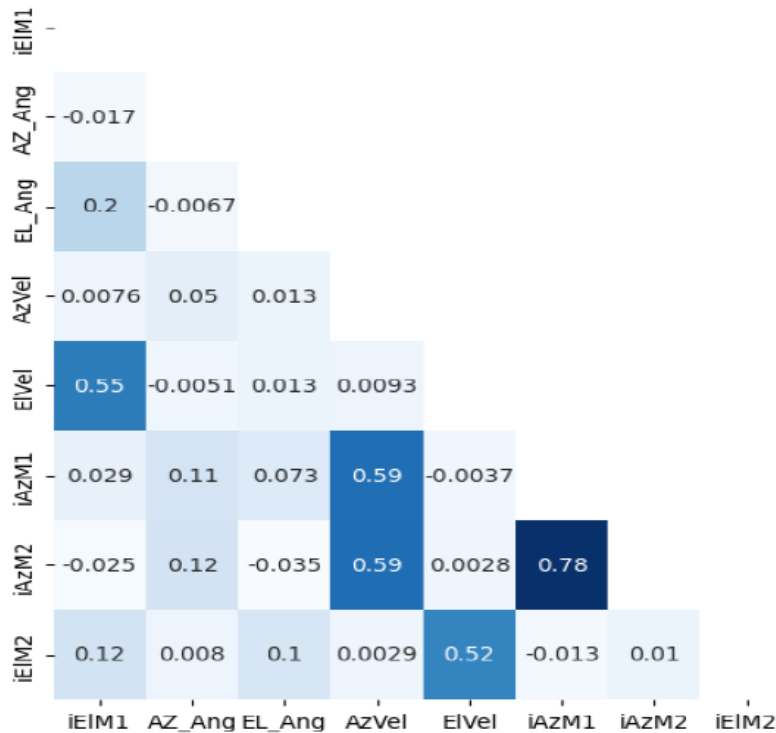


Fig.1. Correlation matrix of important features in the dataset.

Modelling:

The modeling section of this project focuses on developing accurate prediction models for the equipment performance of the Antenna Control Servo System (ACSS). Leveraging machine learning techniques, particularly Long Short-Term Memory (LSTM) networks, the objective is to forecast future ACSS behavior based on historical telemetry data. The modeling process involves several key steps, including data preprocessing, feature selection, model architecture design, training, and evaluation. Through comprehensive data preprocessing, relevant features are extracted and prepared for input into the LSTM network. The LSTM architecture is chosen for its ability to capture temporal dependencies and patterns in sequential data, making it well-suited for time-series prediction tasks like ACSS performance forecasting. Once trained, the LSTM model undergoes rigorous evaluation using appropriate metrics to assess its accuracy, robustness, and generalization capability.

The model architecture consists of multiple layers, each serving a specific purpose in processing the input data and extracting relevant features. It begins with an input layer, followed by two LSTM layers. The LSTM layers are configured to return sequences to capture temporal dependencies effectively. Subsequently, two dense layers are added for feature extraction and prediction, with the final layer using a linear activation function for regression tasks. The model is designed to learn from the sequential data of the Antenna Control Servo System (ACSS) and predict future performance based on historical telemetry data. Through this layered approach, the model can effectively capture complex patterns and dependencies in the sequential data, enabling accurate performance prediction and proactive maintenance strategies. Figure 2 represents the architecture of the LSTM model employed.

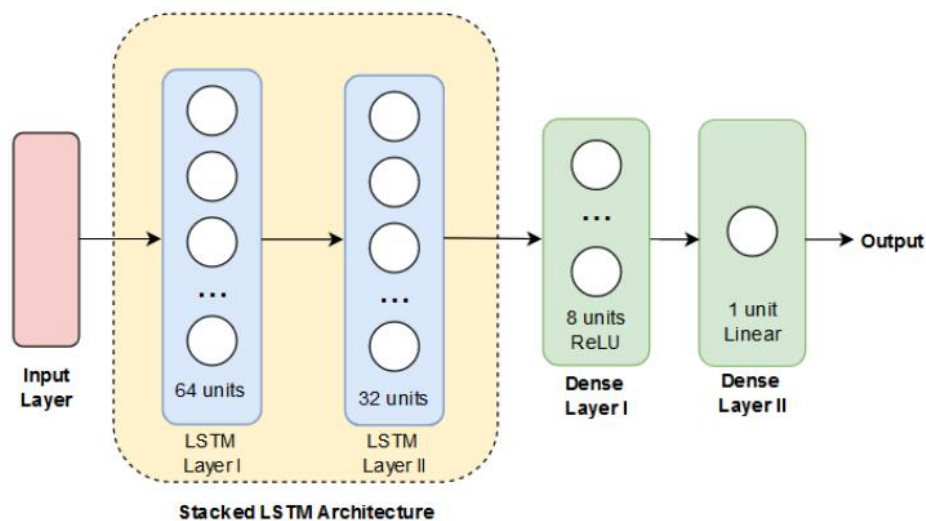


Fig.2. The architecture of the LSTM model employed.

2. Visibility File Prediction of Satellite:

Data Pre-processing and Analysis:

To prepare the data for modelling, several pre-processing steps were undertaken. Initially, the discontinuous dataset was transformed into a continuous one by appending all passes in the visibility file. Additionally, timestamps where there were no passes were filled with 0s to maintain continuity in the dataset. This ensured that the dataset covered the entire duration of interest, from November 2022 to March 2024. The dataset contained parameters such as Azimuth rate (AZ-RATE), Elevation rate (EL-RATE), Azimuth angle (AZI), Elevation angle (ELV), Range (RANGE), Height (HT) etc. The data was discontinuous, containing samples only when the satellite was visible to the ground station. The below figures illustrate the variation of ELV and AZ-RATE throughout the dataset.

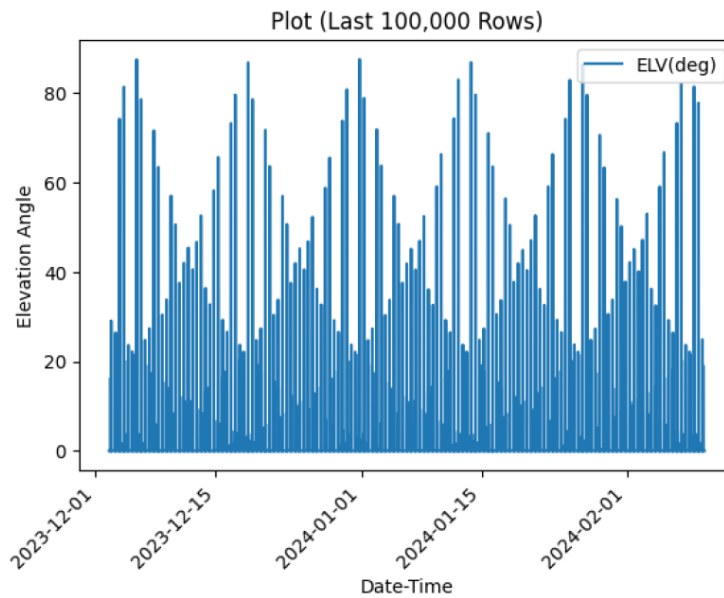


Fig.3. ELV variation in the dataset.

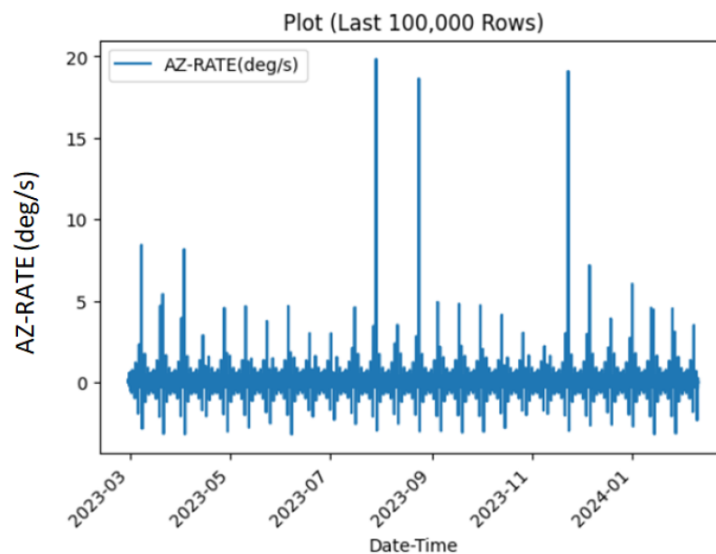


Fig.3. AZ-RATE variation in the dataset

Modelling:

Several modelling approaches were considered to predict satellite visibility, starting with traditional statistical time series models such as ARIMA, VARMA, and VARIMA. Since, this problem was not of a traditional time series nature, the above statistical models failed. Hence zeroes were inserted in timestamps to make it continuous. However, these models failed to provide satisfactory results due to the presence of 0s in the dataset, which affected the moving average component. Consequently, advanced deep learning models capable of handling sequence prediction tasks were explored. Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks are specialized deep learning architectures adept at handling sequential data, making them ideal for tasks like satellite visibility prediction. RNNs maintain internal states allowing them to capture temporal dependencies, but they struggle with long-term memory due to vanishing gradients. LSTMs address this by incorporating memory cells and gates that regulate information flow, enabling them to selectively retain or forget information over time. This unique architecture empowers LSTMs to effectively capture both short-term and long-term dependencies in the data, making them superior for time series forecasting tasks. In the context of satellite visibility prediction, LSTMs excel at learning complex patterns and relationships, facilitating accurate predictions of future visibility parameters based on past observations. The proposed model can be used for other satellite visibility prediction.

In the modeling section, a deep learning architecture comprising stacked LSTM layers with dropout regularization followed by dense layers was utilized. The model begins with an input layer configured to accept sequences of 10 time steps and 1 feature per step. Stacked LSTM layers, with 64 and 32 units respectively, were employed to capture temporal dependencies in the satellite visibility data, with the first LSTM layer returning sequences to feed into the subsequent layers. Dropout layers with rates of 0.2 and 0.1 were incorporated to mitigate overfitting. Following the LSTM layers, two dense layers with 8 and 1 units respectively, equipped with ReLU and linear activation functions, were employed for further processing and output generation. This architecture was chosen for its ability to effectively capture sequential patterns and relationships in the data, thereby enabling accurate predictions of future satellite visibility parameters. Figure 5 represents the architecture of the model employed.

Layer (type)	Output Shape	Param #
lstm (LSTM)	(None, 10, 64)	16,896
dropout (Dropout)	(None, 10, 64)	0
lstm_1 (LSTM)	(None, 32)	12,416
dropout_1 (Dropout)	(None, 32)	0
dense (Dense)	(None, 8)	264
dense_1 (Dense)	(None, 1)	9

Fig.5. The architecture of the LSTM model employed.

3. Trend Analysis of ACU Parameters:

Data Pre-processing and Analysis:

Data collection involved gathering servo log data from Bangalore based ground station for various satellite passes. Pre-processing steps included data cleaning and feature extraction to prepare the dataset for analysis. The data was processed to extract relevant information, such as the satellite name (SAT), date and day of the satellite pass, which was then utilized to calculate the Day of Year (DOY), Pass of the day (POD) and total seconds for each record in the data frame. Figure 6 shows the different features present in the dataset.

	HH	MM	SS	MS	AZ_Ang	EL_Ang	AZPerr	ELPerr	MRxAz	MRxEL	...	CDTmm	CDTss	CDTms	DOY	YY	Seconds	SAT	SAT_NO
0	0	17	10	0	30.78	1.75	0.00	0.0	-1.77	3.50	...	0.0	0.0	0.0	182	2022	1656721030	E04	31
1	0	17	11	0	30.79	1.75	0.05	0.0	-2.69	3.76	...	0.0	0.0	0.0	182	2022	1656721031	E04	31
2	0	17	12	0	30.87	1.75	0.04	0.0	-3.49	3.85	...	0.0	0.0	0.0	182	2022	1656721032	E04	31
3	0	17	13	0	30.95	1.75	0.03	0.0	-3.61	3.68	...	0.0	0.0	0.0	182	2022	1656721033	E04	31
4	0	17	14	0	31.03	1.75	0.02	0.0	-3.65	3.44	...	0.0	0.0	0.0	182	2022	1656721034	E04	31
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
4634795	6	7	17	0	-149.54	2.00	-0.01	0.0	-0.02	-0.01	...	0.0	0.0	0.0	30	2024	1706681237	R2A	41
4634796	6	7	18	0	-149.58	2.00	0.00	0.0	-0.02	-0.01	...	0.0	0.0	0.0	30	2024	1706681238	R2A	41
4634797	6	7	19	0	-149.63	2.00	0.01	0.0	-0.02	-0.01	...	0.0	0.0	0.0	30	2024	1706681239	R2A	41
4634798	6	7	20	0	-149.66	2.00	-0.01	0.0	-0.02	-0.01	...	0.0	0.0	0.0	30	2024	1706681240	R2A	41
4634799	6	7	21	0	-149.68	2.00	0.00	0.0	-0.02	-0.01	...	0.0	0.0	0.0	30	2024	1706681241	R2A	41

4634800 rows x 50 columns

Fig.6. Different features present in the dataset.

Analysis and Observations:

The below graph depicts the average value of current drawn weekly from 2022 to 2024 (72 weeks) from Elevation Motor 1 for all the satellites from Bangalore based ground station.

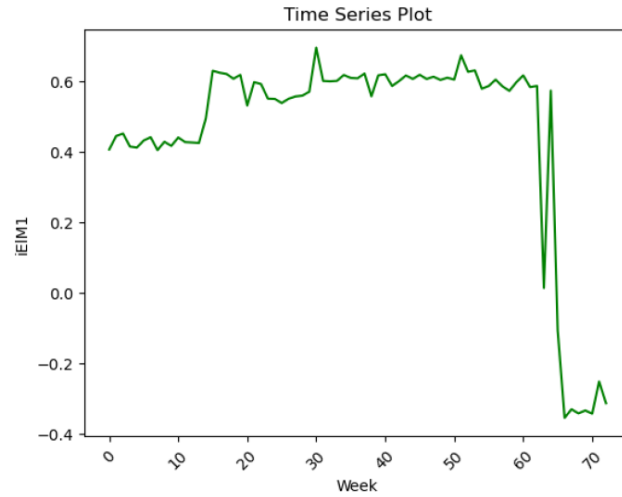


Fig.7 Variation of iELM1

Observations:

- Observed a rise in iELM1 from 0.4 to 0.6 in the weeks 10-15.
- Observed a drastic drop in iELM1 in Week 61 and again a sharp rise in iELM1 in Week 64.
- Observed a drastic drop in iELM1 from 0.5 to -0.3 in Week 65.

The below graph depicts the average value of current drawn weekly from 2022 to 2024 from Elevation Motor 2 for all the satellites from Bangalore based ground station.

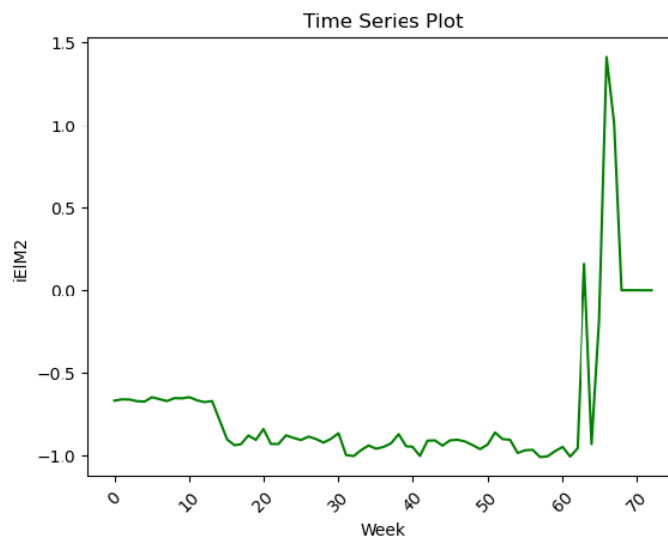


Fig.8 Variation of iELM2

Observations:

- Observed a drop in iELM2 from -0.6 to -1.0 in the weeks 10-15.
- Observed a sharp rise in iELM2 in Week 61 and again a drastic drop in iELM2 in Week 64.
- Observed a drastic rise in iELM2 in Week 65.
- Observed a drastic drop in iELM2 in Week 70 (12th December 2023) and continued with 0 from then on.

The below graph depicts the amount of current drawn (absolute value) from Elevation Motor 1 for 1° change in Elevation Angle from 2022 to 2024 for different ranges of maximum Elevation Angle as depicted by different colours for X2 satellite.

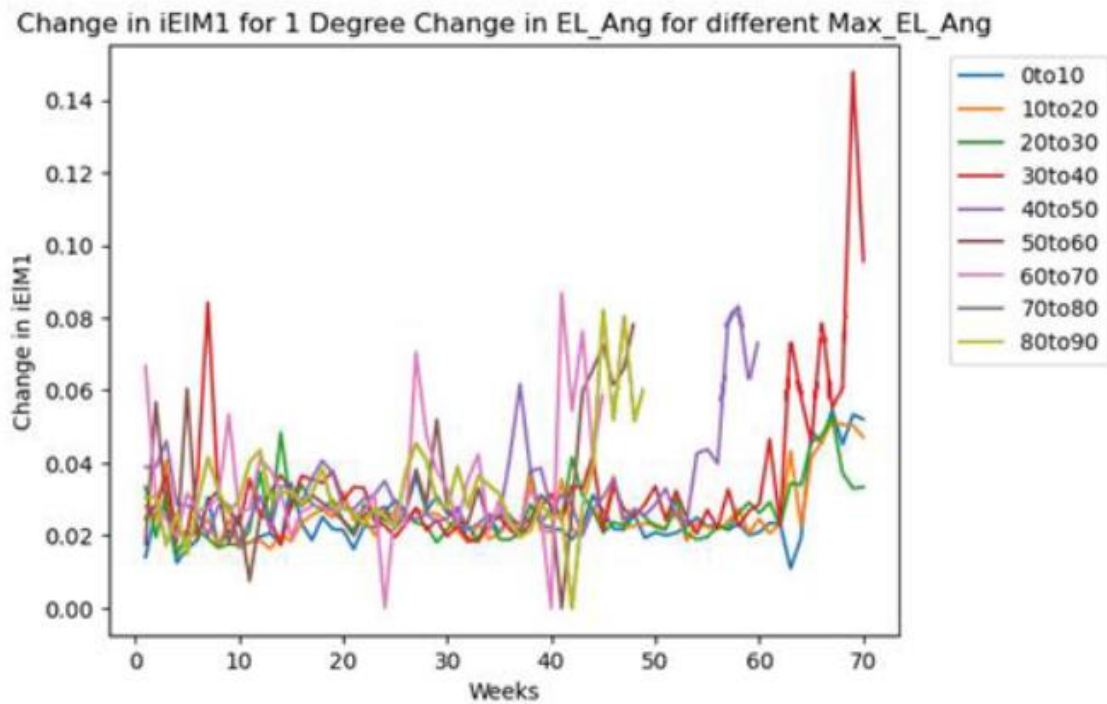


Fig.9. Change in iELM1 for 1° change in EL_Ang for different ranges

Observations:

- Observed a change (increase) in iELM1 for 1° change in El_Ang after Week 60.

The below graph depicts the amount of current drawn (absolute value) from Elevation Motor 2 for 1° change in Elevation Angle from 2022 to 2024 for different ranges of maximum Elevation Angle as depicted by different colours for X2 satellite.

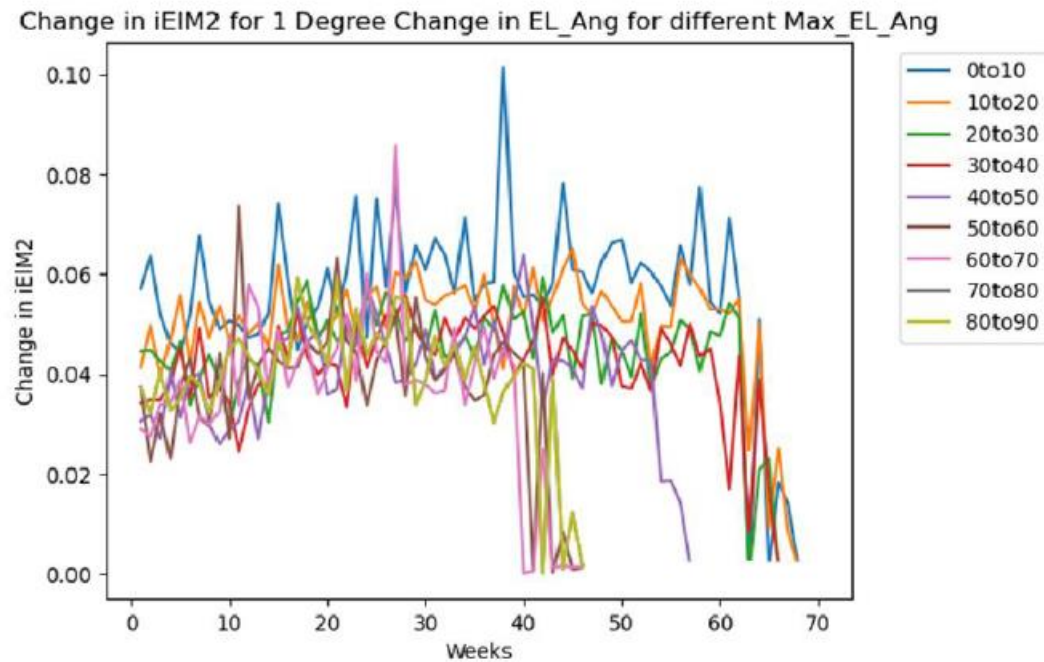


Fig.10. Change in iEIM2 for 1° change in EL_Ang for different ranges

Observations:

- Observed a change (decrease) in iEIM2 for 1° change in EL_Ang after Week 60.

The below graphs depict the relationship between Elevation Motor 1 Current (iEIM1) and Elevation Motor 2 Current (iEIM2) vs Elevation Motor Torque Demand (TelDm) and the sum of both Elevation Motor 1 Current (iEIM1) and Elevation Motor 2 Current (iEIM2) vs Elevation Motor Torque Demand (TelDm) respectively for a pass of the X2 satellite on the first day in the dataset i.e is 1st July 2022.

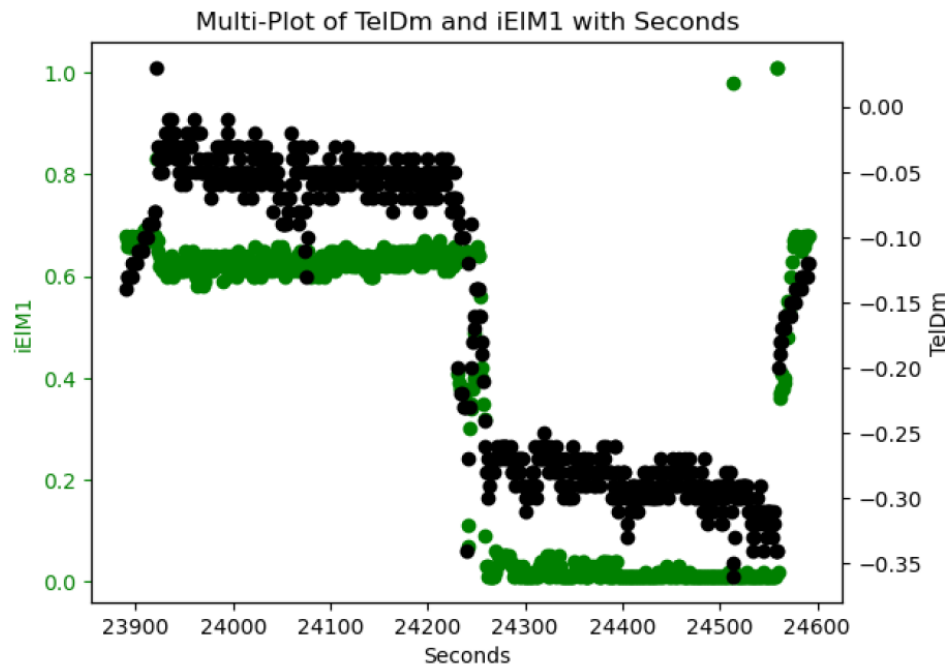


Fig.11. iEIM1 vs TelDm

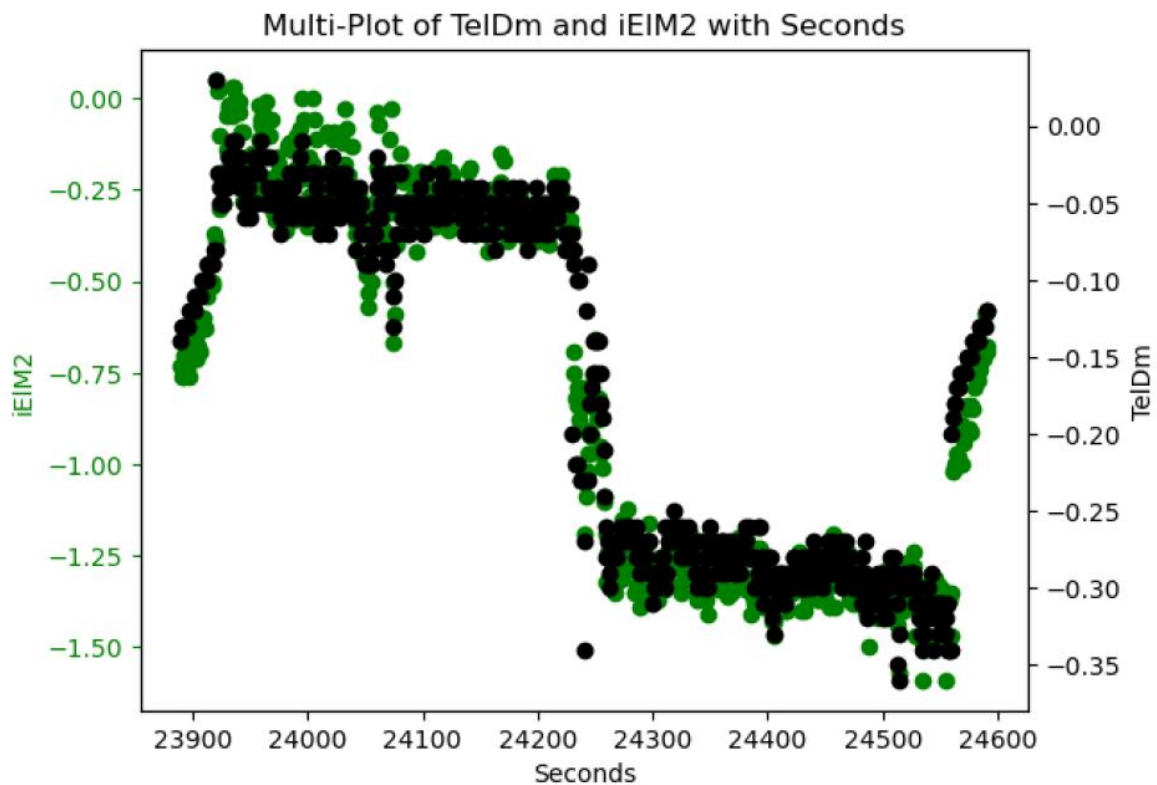


Fig.12. iEIM2 vs TelDm

The below graph depicts the relationship between Elevation Angle (EL_Ang) vs Elevation Motor 1 Current (iEIM1) vs Elevation Motor 2 Current (iEIM2) vs Mode of Operation (Mode) vs ELPerr (Elevation Program Error) for a failure pass case in which the Elevation Velocity (ElVel) has gone below -9 deg/s.

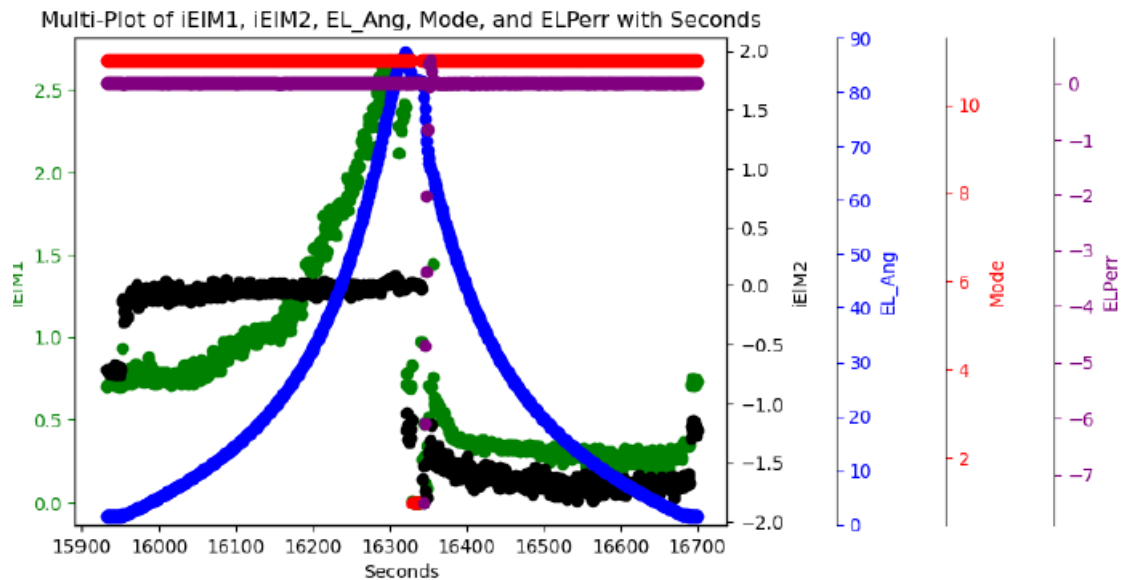


Fig.13. Multi-Plot of iEIM1 vs iEIM2 vs EL_Ang vs Mode vs ELPerr

The below graph depicts the relationship between Elevation Angle (EL_Ang) vs Elevation Motor 1 Current (iEIM1) vs Elevation Motor 2 Current (iEIM2) vs Mode of Operation (Mode) vs ELPerr (Elevation Program Error) for a failure pass case in which the Elevation Velocity (ElVel) has gone above +9 deg/s.

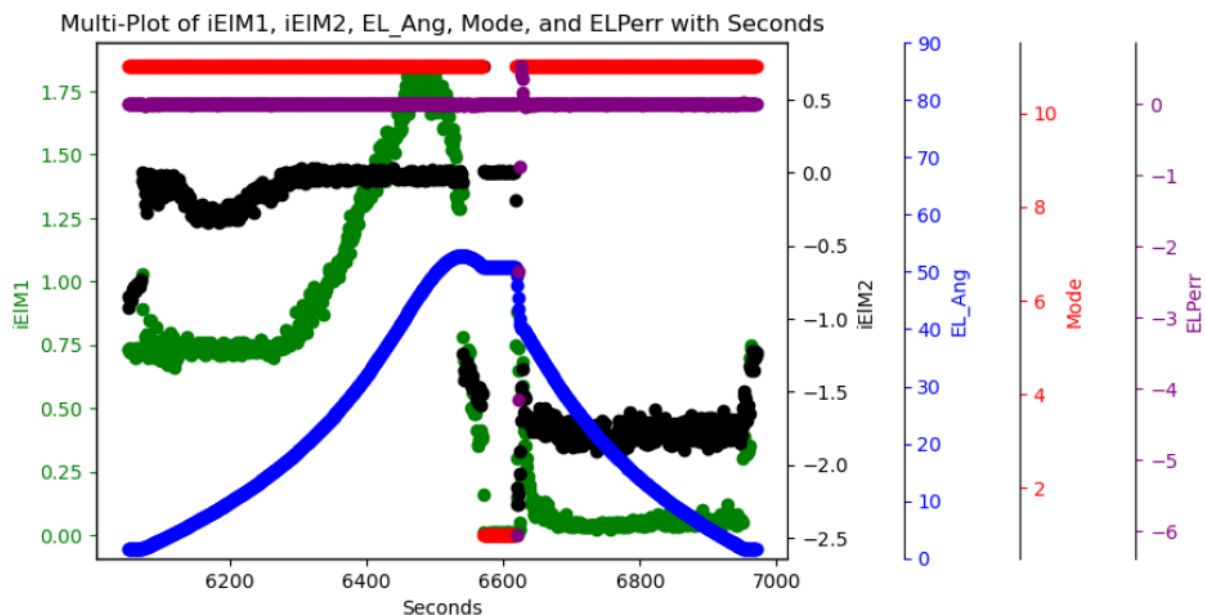


Fig.14. Multi-Plot of iEIM1 vs iEIM2 vs EL_Ang vs Mode vs ELPerr

CHAPTER 7

REFLECTION AND ANALYSIS

During the course of this project, I gained invaluable insights into data handling techniques through the manipulation of vast and intricate sensor data sets. My understanding expanded significantly as I delved into preprocessing methods such as data standardization, normalization, and the conversion of discontinuous data into continuous formats. These techniques proved essential in preparing the data for analysis and modeling, allowing me to effectively manage the complexities inherent in large-scale data sets.

Furthermore, my exploration into time series forecasting introduced me to various models, including RNN (Recurrent Neural Networks), LSTM (Long Short-Term Memory), and GRU (Gated Recurrent Units). Implementing these models for forecasting ACU (Antenna Control Unit) current draws provided valuable insights into the predictive capabilities of these advanced algorithms, highlighting their potential for enhancing forecasting accuracy in complex systems.

In addition to honing my skills in data handling and modeling, I developed a keen understanding of data analysis and visualization through trend analysis on various ACU parameters. Analyzing trends over time allowed me to discern patterns, identify anomalies, and extract meaningful insights from the data. Utilizing visualization techniques facilitated the communication of findings and provided clarity in understanding complex data relationships.

Moreover, my engagement with ISRO's missions and space technology proved to be an enriching experience. Interacting with scientists and experts provided me with firsthand knowledge of missions such as Chandrayaan and Gaganyaan, offering a deeper understanding of ISRO's contributions to space exploration. Learning about ISRO's endeavors and technological advancements broadened my perspective and underscored the importance of collaboration and innovation in advancing space exploration efforts.

In conclusion, this project not only provided me with technical expertise in data handling, modeling, and analysis but also offered a unique opportunity to immerse myself in the fascinating world of space technology. The exposure to real-world data sets and interactions with experts at ISRO have been instrumental in my professional development, equipping me with valuable skills and insights that will undoubtedly shape my future endeavors in data science and space exploration.

CHAPTER 8

RECOMMENDATIONS

- 1. Develop Structured Programs:** Implement structured internship programs with clear learning objectives, project assignments, and mentorship opportunities. Design programs with defined milestones and deliverables, providing interns with a roadmap for their learning and development. Mix classroom training, hands-on projects, and mentorship sessions to maximize learning and skill development.
- 2. Foster Collaboration with Institutions:** Collaborate with educational institutions to involve students in research projects, competitions, and workshops. Establish partnerships to offer joint internship programs, research collaborations, and academic exchanges. Engage with faculty members and research centers to tap into expertise and resources, enriching the internship experience and fostering closer ties between academia and industry.
- 3. Provide Mentorship and Support:** Offer mentorship and guidance to interns, pairing them with experienced professionals who can provide advice, feedback, and support. Mentors can help interns navigate challenges, develop new skills, and achieve their goals. Regular check-in meetings, mentorship workshops, and networking opportunities can enhance the mentorship experience.
- 4. Encourage Innovation and Creativity:** Create an environment that encourages interns to explore innovative ideas and contribute to ongoing projects. Empower interns with autonomy and resources to pursue their interests, participate in brainstorming sessions, hackathons, and design challenges to stimulate creativity and inspire new solutions to complex problems.
- 5. Facilitate Networking Opportunities:** Organize networking events and activities to enable interns to connect with professionals from various departments. Plan meet-and-greet sessions, panel discussions, and industry tours where interns can interact with employees, alumni, and industry experts. Establish online networking platforms to facilitate ongoing communication and collaboration.

CHAPTER 9

CONCLUSION

In conclusion, this project addresses the critical need for proactive maintenance strategies in unmanned satellite tracking ground stations, particularly predictive maintenance through equipment performance monitoring focussing on Antenna Control Servo Systems (ACSS). The methodology employed in this project involved the development of a predictive maintenance system utilizing deep learning techniques, specifically Long Short-Term Memory (LSTM) networks. Through comprehensive feature selection and pre-processing of historical servo log data, the dataset was prepared for modelling, with stacked LSTM networks selected to capture complex temporal patterns and dependencies. The results obtained from the predictive maintenance system demonstrate promising outcomes, with the model effectively predicting future equipment performance or current drawn for future satellite passes based on current drawn during historical satellite passes. Metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R2 value indicate the accuracy of the model predictions. Overall, the successful implementation of the predictive maintenance system underscores the potential of deep learning techniques in enhancing the reliability and efficiency of unmanned ground station operations. Moving forward, future research could explore additional environmental factors and further optimize the predictive maintenance system to address evolving operational needs and challenges.

This project also showcases the effectiveness of LSTM networks in predicting satellite visibility based on historical data. By transforming the discontinuous dataset into a continuous one by appending 0s for timestamps without any satellite passes and employing LSTM for sequence prediction, accurate forecasts of satellite visibility parameters were achieved. The findings highlight the potential of deep learning models in enhancing the capabilities of satellite tracking and communication systems, paving the way for improved operational efficiency and performance.

The trend analysis of antenna control unit servo log data yielded valuable insights into the performance characteristics and operational behaviour of the system during satellite passes. By identifying trends, correlations, and anomalies in various parameters, this analysis contributes to enhancing the reliability, efficiency, and maintenance of satellite communication systems.

CHAPTER 10

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